Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

| **Title of Proposed Technology** | Assured Neuro-Symbolic Reinforcement Learning for Resilient Autonomous and Robotic Systems | | | |
|---|---|---|---|---|
| **Korea's 12 National Strategic Technology Fields** | Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ),<br>Advanced Bio ( ), Aerospace/Marine (√ ), Hydrogen ( ), Cybersecurity (√ ), AI (√),<br>Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( ) | | | |
| **Industrial Technology Classification** [1)] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Machinery and materials* | | Information and Communication | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Robotics/Automation Machinery* | *Robot Control, Intelligence and Autonomous Systems* | Software/AI Systems | AI, Learning and Decision-Making Algorithms |
| **Concept of Proposed Technology** | The proposed technology focuses on the development of **assured neuro-symbolic reinforcement learning (RL) frameworks** for autonomous and robotic systems operating in uncertain, dynamic, and adversarial environments. The core concept is to integrate data-driven learning (deep reinforcement learning) with symbolic reasoning and formal methods to guarantee safety, robustness, | | | |

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| | and predictability of autonomous decision-making.<br>Key capabilities include:<br>• Neuro-symbolic learning that combines neural networks with formal task and safety specifications<br>• Real-time mission monitoring using temporal logic–based reasoning<br>• Safety- and security-aware reinforcement learning resilient to disturbances and cyber-physical attacks<br>• Scalable multi-agent coordination with compositional stability guarantees<br>This technology is applicable to advanced robotics, autonomous mobility platforms, industrial automation, and safety-critical cyber-physical systems. |
| **Rationale for Joint R&D Project** | *Joint R&D is essential to translate advanced assured autonomy algorithms into deployable robotic and industrial systems. While the research institute contributes cutting-edge AI, control, and assurance methodologies, collaboration with Korean SMEs is required to integrate these methods into real products and operational platforms.*<br>***Rationale for collaboration:***<br>• *Research institute provides expertise in neuro-symbolic AI, reinforcement learning, and formal assurance*<br>• *Korean SME partner provides system engineering, embedded implementation, and industrial validation*<br>• *Co-design of algorithms and hardware/software platforms is required to meet reliability, safety, and certification requirements*<br>***Expected benefits of joint R&D:***<br>• *Reduced technical and certification risk for AI-enabled autonomous systems*<br>• *Faster transition from research prototypes to industrial-grade solutions*<br>• *Creation of exportable, high-value autonomous and robotic technologies* |
| **Key Words** | • Assured Autonomy<br>• Neuro-Symbolic Reinforcement Learning<br>• Safety-Critical AI<br>• Multi-Agent Autonomous Systems<br>• Resilient Robotics |

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## 3. Research and Development Objectives & Plan

| **Technology Readiness Level** (TRL) [2] | | **Start of the Project** | Level (3) – Laboratory-scale validation | **End of the Project** | Level (6) – Prototype demonstration in relevant environments |
|---|---|---|---|---|---|
| **R&D Objectives** | | *The objective of this R&D project is to develop assured AI-enabled autonomy technologies that enable robotic and autonomous systems to operate safely, reliably, and predictably in complex environments.*<br>*Specific objectives include:*<br>• *Development of neuro-symbolic reinforcement learning algorithms with formal safety and mission guarantees*<br>• *Integration of temporal-logic-based task specifications into learning and control loops*<br>• *Development of resilient learning mechanisms robust to disturbances, faults, and cyber attacks*<br>• *Demonstration of scalable multi-agent autonomy with provable stability and performance* | | | |
| **R&D Target Specifications** | | • ***Safety Assurance:*** *100% compliance with predefined safety constraints during operation*<br>• ***Robustness:*** *Maintain stable performance under at least 30% sensor noise or communication degradation*<br>• ***Learning Efficiency:*** *Reduce training time by at least 40% compared to baseline deep RL methods*<br>• ***Multi-Agent Scalability:*** *Demonstrate coordinated control of 5–10 autonomous agents with guaranteed stability* | | | |
| **R&D Plan** | | **Year 1: Algorithm Design and Simulation Validation**<br>• Develop neuro-symbolic RL architectures with temporal logic specifications<br>• Implement safety- and security-aware learning algorithms<br>• Validate performance through high-fidelity simulations<br>**Year 2: Prototype Integration and Laboratory Testing**<br>• Integrate algorithms into robotic or autonomous system platforms<br>• Conduct laboratory-scale experiments under uncertainty and disturbances<br>• Refine assurance metrics and learning efficiency<br>**Year 3: System Demonstration and Industrial Validation**<br>• Demonstrate prototype systems in realistic operational scenarios<br>• Evaluate robustness, safety, and scalability<br>• Prepare technology for transfer and commercialization | | | |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | • *Development of core neuro-symbolic reinforcement learning and control algorithms*<br>• *Design of formal specification, safety assurance, and verification frameworks*<br>• *Simulation-based validation and performance analysis*<br>• *Technical leadership and training support for the industrial partner* |
| **Expected R&R of Korean SME Partner** | • *Development and integration of hardware and embedded software platforms*<br>• *System-level implementation of AI and control algorithms*<br>• *Experimental testing, data collection, and validation*<br>• *Contribution to productization, certification, and commercialization strategies* |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | *The joint R&D project is expected to deliver assured AI-enabled autonomy technologies validated on real robotic or autonomous platforms. Technologically, the project will advance the state of the art in safe and resilient reinforcement learning, bridging the gap between academic AI research and industrial deployment.*<br>*From an industrial perspective, Korean SMEs will gain access to high-value assured autonomy technologies applicable to advanced robotics, autonomous mobility, aerospace, and smart manufacturing. The project will strengthen international collaboration, enhance global competitiveness, and support the development of trustworthy AI-driven products.* |

## 6. References

(1) A. Kanellopoulos, L. Zhai, F. Fotiadis, **K. G. Vamvoudakis**, *Control and Game Theoretic Methods for Cyber-Physical Security*, Elsevier, 2024.

(2) J. Netter, G. P. Kontoudis, **K. G. Vamvoudakis**, "Decentralized Multi-Agent Motion Planning using Cognitive Hierarchy and Gaussian Process Classification," to appear in, *IEEE Transactions on Intelligent Vehicles*, 2026.

(3) N.-M. T. Kokolakis, **K. G. Vamvoudakis**, W. M. Haddad, "Fixed-Time Learning for Safe Time-Critical Verification using Reachability Analysis," *Automatica*, vol. 183, paper 112528, 2026.

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| **Description** | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |